

A NOTE ON SAMPLING DISTRIBUTIONS OF HIGHER ORDER MOMENTS*

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1. Introduction and summary

In this note it is shown that the sampling distributions of the higher order factorial and power moments, when sampling from discrete populations, are of the univariable multinomial type.

The method of derivation comprises the formation of the probability generating function (pgf) of the moment concerned with respect to the probability function (pf) from which the samples are drawn. Although trivial to prove, the results are interesting. In the case of the second order moment, the results confirm a conjecture made by Church (1926, p. 389) in his empirical study of the sampling distribution of the variance, when sampling from a binomial population.

2. The sampling distribution

Let $x_i, i=1, \dots, n$, be a random sample of n observations of a discrete random variable X with pf

$$(1) \quad p(x) = P(X = x), \sum_x p(x) = 1, x = 0, 1, \dots, r, \dots,$$

where X may be defined over a finite (r) or countably infinite range of values.

The pgf of the factorial moment, $M'_s(s)$, of order s , defined by

$$M'_s(s) = \frac{1}{n!} \sum_{i=1}^n X_i^{(s)}, (X_i = 0, 1, \dots, r, \dots; s = 1, 2, \dots),$$

is given by (cf. Aitken, 1945, p. 16)

$$(2) \quad P(t) = \prod_{i=1}^n \left\{ \sum_{x_i} p(x_i) t^{x_i} \right\}^{(s)/n},$$

where $y^{(s)} = y(y-1) \dots (y-s+1)$ and the summation is performed over all the values of x_i for which the pf $p(x)$ is defined.

Performing the summation within the brackets and remembering that $y^{(s)} = 0$ for $y < s-1$, equation (2) may be written as

$$(3) \quad P(t) = \left\{ \left(\sum_{x=0}^{s-1} p(x) \right) + p(s)t^{s/n} + \dots + p(r)t^{r/n} + \dots \right\}^n$$

*This work is part of a project supported by the Council for Scientific and Industrial Research, Pretoria.

This may be conveniently written as

$$(4) \quad P(t) = \left(\sum_x p(x) t^j \right)^n; (j = x^{(s)}/n, x = 0, 1, \dots, s, \dots, r, \dots),$$

where $p(x)$ is the given pf and the summation is performed over all values of x for which $p(x)$ is defined. The probability $P(M'_{(s)} = m)$ of obtaining a given value m for the factorial moment of order s will be given by the coefficient of t^m in the expansion of equation (4) in powers of t .

Similarly the pgf of the power moment M'_s , of order s , about the origin, defined by

$$M'_s = \frac{1}{n} \sum_i X_i^s, (X_i = 0, 1, \dots, r, \dots; s = 1, 2, \dots)$$

is given by

$$(5) \quad P(t) = \left(\sum_x p(x) t^{x^s/n} \right)^n,$$

where the summation is defined as for equation (4).

Both equations (4) and (5) represent the pgf's of a univariable multinomial pf. Hence the sampling distributions of both the factorial and power moments are of the univariable multinomial type.

3. Sampling from a binomial population

Consider samples of size n from a binomial population with parameters r, p . The substitution of $p(x) = \binom{r}{x} p^x q^{r-x}$, ($x=0, 1, \dots, r$) in equations (4) and (5) produces the pgf's of the corresponding sampling distributions. The following cases may be noted.

For $s = 1$ both equations (4) and (5), when sampling from a binomial population, reduces to the well-known result

$$(6) \quad P(t) = (q + pt^{1/n})^{rn},$$

being the pgf of the binomial distribution arrayed by $(q + p)^{nr}$, the interval-being $1/n$ instead of unity.

For $2 \leq s \leq r$ the sampling distribution of the factorial moment $M'_{(s)}$ generated by equation (4) has been studied in detail for several values of r, p and n . Except for $s=r$, which will be considered separately, the sampling distributions are multimodal in form for all values of p and small values of n . These are positively J-shaped with one or more peaks at the right-hand tail-ends for the smaller values of p , changing to skew and multimodal forms for larger values of p ; with the number of peaks increasing as p increases. As p approaches unity the form of the distribution becomes negatively J-shaped with two or more peaks at the left-hand tail-ends. For increasing n this pattern of different forms remains more or less the same, except that for $0.6 < p \leq 0.8$ the distribution becomes symmetrical. When $n \geq 25$ and $0.6 < p \leq 0.8$ the distributions closely resemble a normal distribution, which is in accordance with a result obtained by Steyn (1956, p. 196).

In the case of $s=r$, equation (4) reduces to

$$P(t) = \{(1-p^r) + p^r t^{r! / n}\}^n,$$

which generates a binomial distribution with parameters n and p^r , the intervals being $r! / n$ instead of unity. This distribution is positively J-shaped for small values of p and moderately large n . It is symmetrical for $p=2^{-1/r}$ (cf. this with $p=1/2$ in the case of the sampling distribution of the mean generated by equation (6)).

Similar results have been obtained for the distribution of the power moments, but unlike the above they have their probability points at unequal intervals.

Sampling from other discrete populations give similar results with their own characteristics.

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Manuscript received 5th June 1969; revised 26th July 1971.